

ABSTRACT

Israel, which forms a unique geographical bridge between three continents, harbors a relatively large number of fungal species due to its wide range of ecosystems. Among the hypogeous fungi, species of the family Tuberaceae have been observed in various habitats, and while some are native species, some are hypothesized to be newly introduced, including edible ones with agricultural commercial potential. Here, we provide a reassessment of the biodiversity of the family Tuberaceae in northern Israel. We report the genus *Reddellomyces* for the first time in Israel, as well as five new species of Tuber in Israel: *T. aff. excavatum*, *T. aff. lusitanicum*, *T. lyonii*, *T. aff. pulchrosporum*, and *T. aff. zambonelliae*. We also provide the assessment of the commercial potential of cultivating *Tuber aestivum* in Israel. The latter was introduced in an oak orchard in northern Israel and was coincidentally found in abundant numbers at a pecan tree (*Carya illinoensis*) grove about 3.5 km away. Beginning in 2013, *T. aestivum* ascocarps were collected seasonally at the site, making it the first reported pecan grove in Israel yielding European summer truffles. This observation has led to the active inoculation of *T. aestivum* mushrooms in a pecan model plot, which has yielded ascocarps as early as 4 years following its establishment. Altogether, the study shows the richness of the family Tuberaceae in northern Israel and further emphasizes the commercial potential of *T. aestivum* and potentially other naturally occurring truffles for intercropping orchards.